Namhyuk **Ahn**

ASSISTANT PROFESSOR AT INHA UNIVERSITY

nmhkahn@gmail.com | nmhkahn.github.io | nmhkahn | Namhyuk Ahn

Research Interests

My research vision is to establish a comprehensive **Generative AI + X ecosystem** that empowers a diverse spectrum of users, from domain experts to the general public. I strive to democratize the benefits of AI by extending them to under-resourced fields, thereby maximizing the societal utility of generative technologies. My work centers on three key pillars: (1) **AX via Gen AI:** Engineering domain-specific adaptations of foundation models for specialized sectors that necessitate high-fidelity, bespoke solutions, (2) **Professional-Grade Gen AI Systems:** Designing workflows that satisfy rigorous professional requirements through fine-grained controllability and steerability, and (3) **Trustworthy Gen AI:** Developing robust frameworks for data security, privacy protection, and ethical alignment to safeguard all stakeholders.

Work Experience

Inha University — *Incheon, Korea*
ASSISTANT PROFESSOR — *Sep. 2024 - Present*
- School of Electrical and Electronic Engineering
- Department of Electrical and Computer Engineering (Graduate School)

NAVER WEBTOON — *Pangyo, Korea*
AI RESEARCHER — *Aug. 2021 - Aug. 2024*
- Built user-centric AI tools designed for cartoon creators/artists (e.g. Cartooner, DreamStyler)
- Researched user privacy-aware Gen AI (e.g. Impasto, which prevents copyright violation by Gen AI models)
- Developed portrait stylization system (e.g. WebtoonMe)

NAVER AI LAB — *Bundang, Korea*
VISITING RESEARCHER — *Sep. 2019 - Oct. 2020*
- Researched data augmentation for image super-resolution (e.g. CutBlur)
- Developed label-efficient conditional generative models
- Co-worked with Jaejun Yoo, Youngjung Uh and Yunjey Choi

NAVER — *Bundang, Korea*
INTERN — *June 2018 - Aug. 2018*
- Developed image-to-image translation pipeline for talking head project

Education

Ajou University — *Suwon, Korea*
PH.D. IN ARTIFICIAL INTELLIGENCE — *Mar. 2016 - Aug. 2021*
- Advisor: Prof. Kyung-Ah Sohn
- Thesis: Toward an Efficient Deep Image Restoration Method

Ajou University — *Suwon, Korea*
BACHELOR OF MEDIA IN DIGITAL MEDIA — *Mar. 2012 - Aug. 2016*

Selected Publications

- Minsuk Ji*, Sanghyeok Lee*, and **Namhyuk Ahn**. Compositional Image Synthesis with Inference-Time Scaling. ICASSP 2026
- Soyoung Yoon, **Namhyuk Ahn**, and In Kyu Park. T2LF: LLM-Guided Multimodal Diffusion for Text-to-Light Field Synthesis. WACV 2026
- Sungnyun Kim, Junsoo Lee, Kibeom Hong, Daesik Kim, and **Namhyuk Ahn**. DiffBlender: Scalable and Composable Multimodal Text-to-Image Diffusion Models. Expert Systems with Applications (ESWA) 2026
- **Namhyuk Ahn**, Wonhyuk Ahn, KiYoon Yoo, Daesik Kim, and Seung-Hun Nam. Imperceptible Protection Against Style Imitation from Diffusion Models. IEEE Transactions on Multimedia (TMM) 2026

- **Namhyuk Ahn**, KiYoon Yoo, Wonhyuk Ahn, Daesik Kim, and Seung-Hun Nam. Nearly Zero-Cost Protection Against Mimicry by Personalized Diffusion Models. **CVPR 2025**
- Jihye Back, **Namhyuk Ahn**, and Jangho Kim. Magnitude Attention-based Dynamic Pruning. **Expert Systems with Applications (ESWA) 2025**
- **Namhyuk Ahn**, Junsoo Lee, Chunggi Lee, Kunhee Kim, Daesik Kim, Seung-Hun Nam, and Kibeom Hong. DreamStyler: Paint by Style Inversion with Text-to-Image Diffusion Models. **AAAI 2024**
- **Namhyuk Ahn**, Jaejun Yoo, and Kyung-Ah Sohn. Data Augmentation for Low-Level Vision: CutBlur and Mixture-of-Augmentation. **International Journal of Computer Vision (IJCV) 2024**
- Jeong-Hyeon Moon*, **Namhyuk Ahn***, and Kyung-Ah Sohn. Decomposing Texture and Semantics for Out-of-distribution Detection. **Expert Systems with Applications (ESWA), 2024**
- **Namhyuk Ahn**, Patrick Kwon, Jihye Back, Kibeom Hong, and Seungkwon Kim. Interactive Cartoonization with Controllable Perceptual Factors. **CVPR 2023**
- **Namhyuk Ahn**, Byungkon Kang, and Kyung-Ah Sohn. Efficient Deep Neural Network for Photo-Realistic Image Super-Resolution. **Pattern Recognition (PR) 2022**
- Jaejun Yoo*, **Namhyuk Ahn***, and Kyung-Ah Sohn. Rethinking Data Augmentation for Image Super-Resolution: A Comprehensive Analysis and a New Strategy. **CVPR 2020**
- **Namhyuk Ahn**, Byungkon Kang, and Kyung-Ah Sohn. Fast, Accurate, and Lightweight Super-Resolution With Cascading Residual Network. **ECCV 2018**

Professional Service

Reviewer CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, AAAI, ACMMM, WACV
TPAMI, IJCV, TIP, TMM, TCSVT, SPIC, ESWA, Neurocomputing

Editor Mathematical Biosciences and Engineering (2022-2023; Guest)

Teaching

2025-2 Deep Learning, Python Programming, Multimedia (English)
2025-1 Multimedia, Vision-Language Models (Graduate), Multimodal AI (Graduate)
2024-2 Computer Graphics, Creative Design for Engineering
2017 Deep Learning and its Applications (at FastCampus)

Awards

2024.1 **NAVER N Innovation Award**, 3rd prize on R&D track
2023.1 **NAVER N Innovation Award**, 2nd prize on R&D track
2018.6 **Honorable Mention Award**, SR Challenge on NTIRE workshops @ CVPR

Invited Talks

2025.10 **ICCE-Asia (Tutorial)**, Responsible Generative AI: From Provenance to Protection
2025.9 **SNU Hospital**, Useful or Harmful? Rethinking Generative AI in Vision
2025.7 **KCC**, Content Protection in Generative AI Era
2023.3 **Ajou University**, Recent Advances in Visual Generative Models

Press Mentions

2024.4 **Digital daily**, "네이버웹툰, '작품 무단 학습 AI' 막는 모델 선보인다"
2023.2 **Tech M**, "미국·유럽·일본도 '감탄'...웹툰 AI 시대 열어야죠"
2023.1 **Digital daily**, "내 얼굴이 웹툰 캐릭터로 변했다"